Today: - few more estimates on hitting probability (exit distribution)

& hitting time (exit time)

(Gambler ruin with $p \neq 1/2$)

x : $E_x[T_x] < \infty$

- connections to recurrence, positive recurrence

Ch. 1

- Branching processes:
  - mean value
  - survival probability.

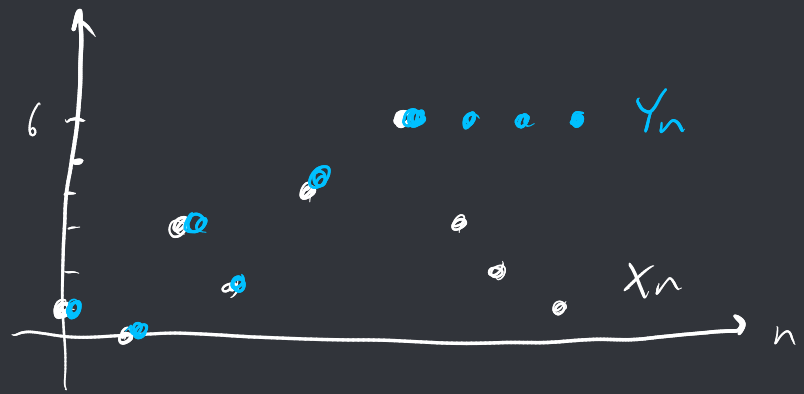
Ch. 2

- Exponential distribution & Poisson process

Stopped process: given (X_n) & stopping time T, process stopped at T is (Y_n) defined by

$$Y_n := X_{n \wedge T}$$

ex: $T = \inf\{n : X_n > 5\}$



h-transform of stopped process $(X_{n \wedge T})_{n \in \mathbb{N}}$ is what (X_n) looks like up to time T conditioned on $X_T \in A$.

It has transition probabilities

$$q(x,y) = p(x,y) \, h(y) / h(x).$$

<u>Proof</u>: $q(x,y) = P(X_{n+1} = y \mid X_n = x, \; X_T \in A, \; T > n)$

use def$^{\underline{n}}$ of cond$^{\underline{l}}$ probab:

$$= \frac{P(X_{n+1}=y,\ X_T \in A \mid X_n = x,\ T>n)}{P(X_T \in A \mid X_n = x,\ T>n)}$$

$$= \frac{P(X_T \in A \mid X_n = x,\ T>n,\ X_{n+1}=y)\,P(X_{n+1}=y \mid X_n = x, T>n)}{P(X_T \in A \mid X_n = x,\ T>n)}$$

Mkv property on both terms

$$= \frac{P(X_T \in A \mid X_{n+1}=y,\ T>n)\,P(X_{n+1}=y \mid X_n = x)}{P(X_T \in A \mid X_n = x,\ T>n)}$$

$$= \frac{h(y)\,p(x,y)}{h(x)}$$

exercise: check that q is a trans. prob:

that $\forall x \quad \sum_y q(x,y) = 1.$

(uses $h(x) = \sum p(x,y)\,h(y)$).

Martingale for Gambler's ruin with $p \neq \frac{1}{2}$.

If $p = \frac{1}{2}$, X_n is a m.g.

If $p \neq \frac{1}{2}$, claim $M_n := \left(\frac{1-p}{p}\right)^{X_n}$ is a m.g.:

compute $E\left[M_{n+1} \mid X_0, \dots, X_n\right]$ .. is it $\overset{?}{=} M_n$

(recall $p(x, x+1) = p$, $p(x, x-1) = 1-p$, $0 < x < N$).

Given $X_0, \dots, X_n$, $M_{n+1} = \begin{cases} \left(\frac{1-p}{p}\right)^{X_{n+1}} & \text{w.p. } p \\ \left(\frac{1-p}{p}\right)^{X_{n-1}} & \text{w.p. } 1-p \end{cases}$

$$\therefore \quad E\left[M_{n+1} \,\middle|\, X_0, \dots, X_n\right] = p \cdot \left(\frac{1-p}{p}\right)^{X_n+1} + (1-p)\left(\frac{1-p}{p}\right)^{X_n-1}$$

$$= \left(\frac{1-p}{p}\right)^{X_n} \underbrace{\left(p \cdot \frac{1-p}{p} + 1-p \, \frac{p}{1-p}\right)}_{\substack{= 1-p+p \\ = 1.}}$$

$$= M_n$$

Using (M_n) & optional stopping we find that

$h(x) = P(X_T = N \mid X_0 = x)$ can be computed :

$$E_x[M_0] = E_x[M_T]$$

$$\left(\frac{1-p}{p}\right)^x = h(x)\left(\frac{1-p}{p}\right)^N + (1-h(x))\cdot\overbrace{\left(\frac{1-p}{p}\right)^0}^{=1}$$

$$\left(\frac{1-p}{p}\right)^x - 1 = h(x)\left(\left(\frac{1-p}{p}\right)^N - 1\right)$$

$$\therefore \quad h(x) = \frac{\left(\frac{1-p}{p}\right)^x - 1}{\left(\frac{1-p}{p}\right)^N - 1} \quad \text{if } p \neq \tfrac{1}{2}.$$

if $p < \frac{1}{2}$, $\frac{1-p}{p} > 1$ so $\left(\frac{1-p}{p}\right)^N$ large if N large.

upper bound:

$$h(x) \le \frac{\left(\frac{1-p}{p}\right)^x}{\left(\frac{1-p}{p}\right)^N} = \left(\frac{1-p}{p}\right)^{x-N}$$

$$\left(\frac{a}{b} \le \frac{a+c}{b+c}, \quad a,b,c>0 \right)$$

If $p = \frac{1}{3}$, $\frac{1-p}{p} = 2$

so for $N = 10$,

$$h(1) \le 2^{-9}$$

exercise: derive "nice" lower bound on $h(x)$ for $p > \frac{1}{2}$.

hitting time of $\{0, N\}$ for G.R., $p \neq \frac{1}{2}$.

<u>Claim:</u> $(*)$ $V_n = X_n - (2p-1)n$ is a m.g.

(generalizes "X_n is a m.g. when $p = \frac{1}{2}$")

<u>Check:</u>
$$E[X_{n+1} \mid X_0, \dots, X_n] = p(X_n + 1) + (1-p)(X_n - 1)$$
$$= X_n + p - (1-p)$$
$$= X_n + 2p - 1.$$

exercise: use this to check $(*)$.

Optional stopping: $E[V_T] = E[V_0]$

$$E_x[X_T - (2p-1)T] = E_x[X_0] = x$$

$$\underbrace{E_x[X_T]} - (2p-1)E_x[T] = x$$

use distribution from previous calculation to compute (get formula for $E_x[T]$)

Just an upper bound: $E_x[X_T] \le N$

$$(2p-1)E_x[T] \le N - x$$

if $p > 1/2$ $\quad E_x[T] \le \dfrac{N-x}{2p-1}$.

☆ <u>Branching process</u>

Given $(p_k)_{k=0}^{\infty}$, $p_k \geq 0$, $\sum_{k=0}^{\infty} p_k = 1$

define $(\xi_{i,n})_{i,n \in \mathbb{N}}$ i.i.d. w/ $P(\xi_{i,n} = k) = p_k$

and let $Z_0 = 1$, $Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{i,n}$.

(Z_n) is a branching process.

<u>Mean value:</u> $E[Z_n] = m^n \quad \forall n$ where

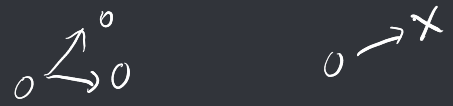
(2)

$$m = \sum_k k p_k$$

$m =$ avg. # of "offspring"

# of individuals that replace each individual in previous gen$^{\underline{n}}$.

ex: if $p_2 = p$ & $p_0 = 1-p$ then $m = 2p$.



if $p > \frac{1}{2}$ population tends to increase

$= \frac{1}{2}$ " constant on average...

check (2): $E[Z_1] = \sum_k k p_k = m$ ✓

if $E[Z_n] = m^n$ then

$$E[Z_{n+1}] = \sum_z \underbrace{E[Z_{n+1} \mid Z_n = z]}_{= E\left[\sum_{i=1}^{z} \xi_{i,n}\right] = \sum_{i=1}^{z} m = mz} P(Z_n = z)$$

$$= \sum_z mz\, P(Z_n = z)$$

$$= m E[Z_n] = m^{n+1}.$$

Define

$$\rho_n = P(Z_n = 0)$$

$$\rho = \lim_{n \to \infty} \rho_n$$

(limit exists because $\{Z_n = 0\} \supset \{Z_{n+1} = 0\}$)

Under what conditions is $\rho < 1$?

$(1 - \rho = P(Z_n \neq 0 \ \forall n))$.

& Continuity of probability says if $A_1 \subset A_2 \subset \cdots$ then $P(\bigcup_n A_n) = \lim_{n \to \infty} P(A_n)$

& if $A_1 \supset A_2 \supset \cdots$ then $P(\bigcap_n A_n) = \lim_{n \to \infty} P(A_n)$

<u>Theorem</u>: if $m > 1$ then $\rho < 1$

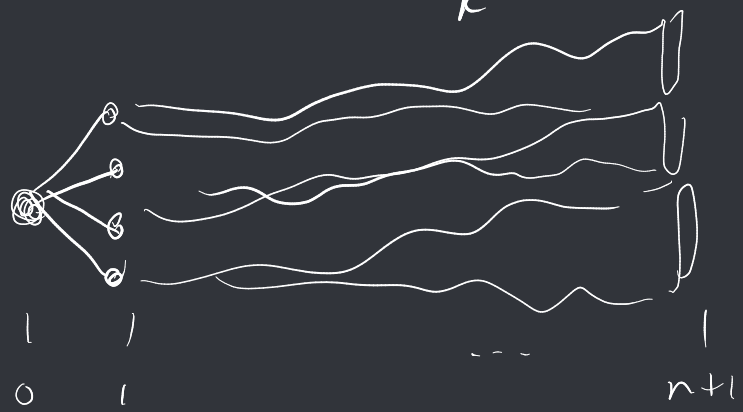
if $m \leq 1$ & $\underbrace{p_1 < 1}_{\text{rules out}}$ then $\rho = 1$.

$\bullet \to \bullet \to \bullet \to \bullet \cdots$

Saw previously that (ρ_n) satisfies

$$\rho_{n+1} = f(\rho_n)$$

where $f(x) = \sum_k p_k x^k$.



generations.

$P(\text{lineage from gen. 1 dead by gen } n+1)$

$= p_n$

each lineage in gen 1 needs to die out by gen. n+1 for whole process to go extinct.

By conditioning on first step,

$$p_{n+1} = P(Z_{n+1}=0) = \sum_k \underbrace{P(Z_{n+1}=0 \mid Z_1=k)}_{= p_n^k} p_k$$

by indep. of distinct lineages.

How to use

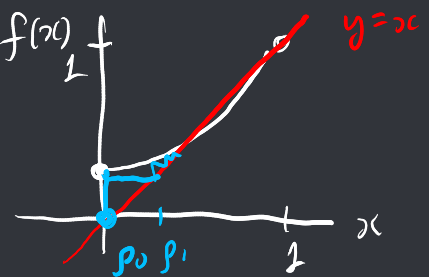
$$p_{n+1} = f(p_n)$$

draw sequence using cobweb diagram:

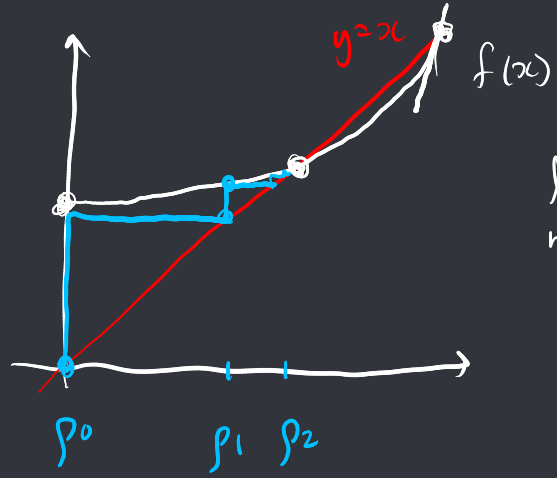
to assess $\lim_{n\to\infty} p_n$?

Study f.

ex: $p_0 = 1-p$, $p_2 = p$; $f(x) = 1-p+px^2$



$p_0 = 0$



$$\lim_{n\to\infty} p_n = \text{smallest value } x : f(x) = x$$

(uses f', $f'' > 0$, $x > 0$)
[check from formula]

$p < 1 \iff f$ has a "fixed point" $x < 1$.

$\iff f'(1) > 1$ (f incr. & convex. & $f(1) = 1$)

However $f'(1) = \sum_k p_k k (1)^{k-1} = \sum_k k p_k = m.$